# Monopath DAGs: Structured Patient Trajectories from Clinical Case Reports

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Clinical machine learning has limited access to rare diseases, atypical responses, and uncommon care pathways, making these scenarios difficult to model or generate synthetically. Case reports document many such trajectories in detail, but free text obscures the patient states, temporal transitions, and decision-relevant information that determine what happens next. We introduce *Monopath Directed Acyclic Graphs (DAGs)*, a framework that converts case reports into ordered sequences of clinician-perceived patient states connected by explicit clinical transitions. We construct Monopath DAGs from 485 lung cancer case reports and first test whether they preserve the source trajectory. Graph-based reconstruction achieves a ClinicalBERT BERTScore of $F_1 = 0.798 \pm 0.051$; clinical review finds 94% node semantic accuracy, 87% correct node ordering, 94% edge correctness, and approximately 98% edge semantic validity. We then test whether preserving this structure changes what models can recover from the data. Graph representations separate clinical trajectories more strongly than text embeddings (Calinski–Harabasz: 110.5 vs. 41.9), and graph-conditioned synthetic cases improve timeline validity ($p = 0.003$), clinical actionability ($p = 0.004$), and most importantly safe clinical actionability ($p < 0.001$) in blinded physician evaluation, with physicians preferring graph-conditioned cases in 62% comparisons. These results show that Monopath DAGs can transform rare, information-rich case reports into structured clinical trajectories that preserve the detail needed for downstream decision-making and improve the quality of synthetic cases generated from them.

*Keywords*: clinical trajectories; directed acyclic graphs; synthetic clinical data; case reports; clinical language models; patient trajectory modeling

## 1. Introduction

Healthcare remains a challenging domain for AI model development. Although clinical AI models have increasingly achieved strong performance on retrospective benchmarks, these gains have failed to consistently translate to prospective clinical evaluation and routine deployment.[1–3] Limited and inconsistent clinical data contribute to this translation gap, particularly for rare diseases and uncommon clinical presentations. As a result, synthetic data generation has become a promising technique for ML practitioners to expand coverage of these underrepresented cases.

Although synthetic data can expand the coverage of rare clinical cases, additional samples do not guarantee the preservation of informative qualities. A generated case may remain coherent and clinically plausible while omitting findings that alter diagnostic reasoning or subsequent actions. Clinical summarization studies have demonstrated this failure: compressed representations can preserve a high-level account of a patient's medical history while excluding information that clinicians need for downstream decisions.[4,5] Synthetic generation therefore depends on preserving information that constrains subsequent clinical decisions, not only representing information sufficient to maintain clinical validity.

Longitudinal EHRs introduce additional loss of representational fidelity across the clinical data pipeline. Clinical encounters encompass temporally-distributed observations, interventions, treatment responses, and outcomes, which clinicians selectively document, and downstream systems subsequently normalize, aggregate, or summarize for analysis. Each transformation can alter the temporal and contextual relationships between events, including treatment-response sequences, unresolved findings, and dependencies between prior observations and subsequent actions. Recent longitudinal EHR studies demonstrate that event-ordering itself carries information associated with later clinical outcomes.[6] Consequently, a representation may retain sufficient information to characterize the current clinical state while failing to preserve the relationships that constrain subsequent clinical decisions.

Current evaluations of synthetic clinical data primarily assess distributional fidelity, task-specific utility, privacy risk, and broad measures of clinical plausibility.[7] These criteria determine whether generated data reproduce statistical properties of observed cohorts, retain performance on predefined prediction tasks, or satisfy general clinical-validity constraints. They do not directly assess whether a synthetic case preserves the sequence of findings, interventions, and responses needed to discriminate between subsequent clinical actions. As a result, two cases may achieve comparable validity or utility scores while materially differing in temporal coherence, actionability, and the safety of the decisions they support.

Narrative case reports provide a useful setting in which to study this gap. Clinicians use case reports to document unusual presentations, diagnostic workups, interventions, treatment responses, and longitudinal disease progression. These reports often retain the sequence of findings, decisions, and responses that condensed EHR representations omit, while disproportionately representing rare or diagnostically challenging cases. However, case reports encode these relationships in free text. Temporal dependencies, decision points, and transitions between clinically-meaningful states often remain implicit within the narrative, limiting systematic extraction and large-scale modeling.

Although case reports may describe competing diagnoses, concurrent findings, and alternative management considerations, they ultimately document a single observed clinical course. Therefore, we can represent each case as a directed path through clinically-meaningful patient states and transitions. We retain this path within a directed acyclic graph (DAG) structure so that individual states and transitions remain explicitly addressable and the representation can support graph operations such as sub-trajectory selection. This study evaluates the complete observed trajectory rather than selected subgraphs, allowing us to test whether explicit trajectory structure preserves information relevant to subsequent clinical decisions.

We present MONOPATH DAGS as structured intermediates for synthetic case generation. The framework extracts clinically-relevant states and transitions from case-report narratives, organizes them into temporally-ordered trajectories, and conditions generation on the resulting graph structure. This design allows us to compare synthetic cases generated with and without explicit trajectory structure while holding the underlying source material constant.

We evaluated this framework on 485 curated lung cancer case reports and 200 rare-disease cases across four disease groups. Four physicians compared graph-conditioned and control-generated cases on clinical validity, timeline validity, actionability, safe actionability, and language appropriateness. This evaluation directly tests whether cases with similar clinical validity can still differ in the information they preserve for subsequent clinical decision-making.

This work makes four key contributions:

- **A decision-relevant evaluation framework** for synthetic clinical data that distinguishes clinical validity from temporal coherence, actionability, and safe actionability.
- **An empirical demonstration that clinical validity alone can mask meaningful differences in downstream decision utility**, showing that representations with similar clinical plausibility can differ in the information they preserve for subsequent clinical reasoning.
- **A structured trajectory representation** that extracts ordered, ontology-grounded Monopath DAGs from free-text case reports and preserves clinically relevant temporal and state transitions.
- **A synthetic data generation pipeline and evaluation dataset** spanning 485 lung cancer case reports and 200 rare disease cases across four disease groups.

## 2. Related Work

**Clinical Narrative Information Extraction.** Prior work in clinical NLP has largely focused on structured sources such as clinical notes, discharge summaries, and radiology reports. Early systems (*e.g.*, cTAKES, MetaMap) combine rules and vocabularies for entity recognition and normalization. Subsequent methods include semi-supervised CRFs for temporal event detection in EMRs[8] and PatternCausality, an unsupervised dependency-based approach to extract cause–effect relations.[9] With pretrained models, transformer-based methods such as ClinicalBERT[10] and BioGPT[11] advance entity and temporal extraction, while GraphTrex achieved state-of-the-art performance on i2b2 TempEval.[12] Case reports have also been targeted: GPT-3 has been used for extraction,[13] and Zhou *et al.* developed CREATe for graph-based cardiovascular case report representations.[14] Complementing this, PMC-Patients released 167,000 case report summaries for retrieval tasks,[15] though the unstructured text limits temporal or causal modeling. In contrast, we transform full case narratives into structured, temporally ordered graphs, enabling reasoning, clustering, and simulation. More broadly, while most modeling centers on structured EHR data, case reports demand methods that account for narrative complexity and rare or atypical presentations—motivating approaches that integrate statistical modeling with clinical knowledge to capture this richness in machine-readable form.

**Patient Trajectory Modeling.** Growing work has focused on structured EHRs and time-series data. Sequence-based models such as RNNs, Transformers, and temporal point processes have been applied to EHR sequences [*e.g.*, RETAIN,[16] BEHRT,[17] Med-BERT[18]]. Graph-based methods have also emerged, treating visits or diagnoses as nodes; for example, GraphCare integrates knowledge graphs while modeling trajectories for clustering and prediction.[19] However, these approaches depend on structured inputs that are not available in free-text case reports, limiting their applicability to narrative clinical data. Methods for extracting trajectories directly from unstructured narrative case reports, essential for studying clinical progression beyond what EHRs capture, remain largely unexplored. Existing approaches typically extract entities or relations without enforcing global temporal structure.

**Synthetic Patient Data Generation.** To address EHR data scarcity and privacy, synthetic generation has gained traction. Rule-based simulators such as Synthea[20] create lifelike trajectories using hard-coded progression logic, while ML-based approaches such as EHR-M-GAN[21] and EHR-Safe[22] leverage GANs and encoder–decoder models to synthesize longitudinal data with privacy safeguards. The HALO model[23] applies hierarchical Transformers to generate disease code sequences, and GPT-style methods have been explored for discharge summary synthesis.[24] These techniques enable robust training without compromising privacy, but struggle to reproduce the rare, atypical trajectories and complex interventions documented in case reports—patterns that are difficult to simulate *de novo*.

Across these works, prior methods either assume structured inputs (EHRs) or extract unstructured outputs (entity spans, summaries). Monopath DAGs enable clinically grounded temporal structure directly on free-text narratives, enabling reuse across clustering, comparison, and generation tasks.

## 3. Methods

## 4. Modular Framework for Patient Trajectory Graph Extraction

We present a modular framework for converting free-text clinical case reports into structured patient trajectories represented as *Monopath Directed Acyclic Graphs (Monopath DAGs).*

### 4.1. *Task Definition*

We represent each clinical case report $d_i \in \mathcal{D}$ as a directed acyclic graph $G_i = (V_i, E_i)$. Each node corresponds to a patient state, defined as a snapshot of the patient as perceived by the clinician at a given point in the trajectory. States reflect the clinical information available at that point rather than an assumed complete representation of the patient's underlying condition. Nodes are annotated with free-text descriptions, structured clinical attributes, and, when available, explicit timestamps. Directed edges represent transitions between states and are labeled by transition type and the clinical entities associated with each transition. See Appendix 1.1 for the complete formal definition of patient states and graph structure.

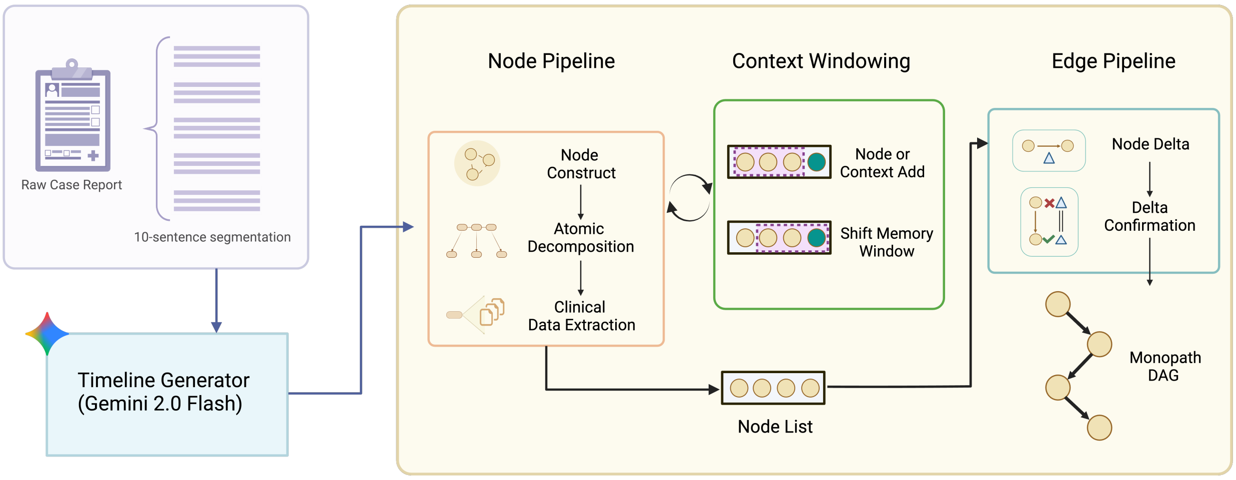



Fig. 1: DAG generation pipeline for patient trajectory modeling.

### 4.2. *Patient Trajectory Generation*

Trajectory construction follows three structural constraints: (i) the graph is acyclic, reflecting the forward progression of clinical time; (ii) each node has at most one incoming edge; and (iii) nodes are temporally ordered using explicit timestamps when available and inferred narrative order otherwise. These constraints define the Monopath representation and reduce ambiguity during trajectory construction.

**Preprocessing & Timeline Extraction.** Each case report is retrieved from PubMed Central as UTF-8 HTML, with non-narrative boilerplate removed. We segment the remaining narrative into 10-sentence blocks and pass them to the timeline generator $T_i = \texttt{PatientTimeline}(p_1, \ldots, p_k)$. We tested Llama 3.1 8B and 70B but found their timeline extraction insufficiently consistent across cases. We therefore selected Gemini 2.0 Flash for more reliable extraction at practical inference speed and cost, using temperature $T = 0.2$ and an approximately 128k-token context window.

**Node Construction.** We divide each patient timeline $T_i$ into four-sentence blocks $c_j$ and process them sequentially to construct patient states. For each block, we generate a node

$$v_j = \{\texttt{node\_id}, \texttt{step\_index}, s_j, a_j, t_j\},$$

where $s_j$ is the free-text state description, $a_j$ contains structured clinical attributes, and $t_j$ is an optional timestamp. To preserve local temporal context, each node $v_j$ is generated using a compressed representation of the three preceding states.

We recursively decompose each state description $s_j$ into atomic clinical facts, defined as statements containing a single clinical assertion about the patient state. We then map these facts to structured ontology terms. Facts that cannot be mapped are retained as free text to preserve information. We then merge overlapping or redundant states and deterministically reindex the resulting sequence.

**Edge Construction.** Given the final node sequence $\{v_1, \ldots, v_n\}$, we construct an edge between each adjacent pair $(v_i, v_{i+1})$ by identifying the clinical change between states. Each edge is represented as

$$e_{i \to i+1} = \{\texttt{edge\_id}, \tau_{ij}, \phi_{ij}\},$$

where $\tau_{ij}$ is a free-text transition description and $\phi_{ij}$ is a structured event containing the trigger type, clinical domain, and associated ontology terms. We use three few-shot demonstrations to promote schema consistency and consistent transition labeling.

**Graph Serialization & Metadata Tracking.** Each patient trajectory is serialized as a JSON directed graph following a consistent schema with ontology-grounded labels and structured metadata (Appendix 2). We enforce schema compliance during construction while deferring full structural validation for downstream evaluation.

### 4.3. *Lung Cancer Case Report Dataset*

We curated a dataset of full-text lung cancer case reports from PubMed Central (PMC). Using the Entrez API, we searched for open-access case reports published from 2019 onward using lung cancer-related terms, including lung cancer, lung carcinoma, non–small cell lung cancer (NSCLC), small cell lung cancer (SCLC), mesothelioma, pulmonary carcinoma, and bronchogenic carcinoma, matched against title and abstract fields. This search yielded 550 case reports.

## 5. Evaluation and Utility

### 5.1. *Graph Construction and Fidelity*

To ensure the LLM- and rule-based transformation with ontology alignment captures a large portion of the semantic and clinical content, we combine direct clinician review with automated fidelity and structural assessments.

**Human Graph Review.** Clinical annotators directly compare each generated graph with its corresponding original case report to evaluate graph fidelity and quality. Reviewers assess whether node and edge preserves the patient states represented in the source case, the clinically relevant information contained within those states, and the temporal ordering of the trajectory (Appendix 4.2).

**LLM-Based Narrative Reconstruction and BERTScore Evaluation.** We reconstruct the input case narrative from the generated graph using Gemini 2.0 Flash. The model receives a payload containing the graph's nodes and edges and is prompted to generate a coherent textual summary that aims to mirror the structure and content of the original case. The reconstructed narrative $\hat{R}$ is compared to the original case report $R$ using BERTScore,[25] which measures semantic similarity via contextualized embeddings from ClinicalBERT. Precision and recall are computed as

$$\text{Precision} = \frac{1}{|\hat{R}|} \sum_{\hat{r} \in \hat{R}} \max_{r \in R} \text{sim}(\hat{r}, r)$$

and

$$\text{Recall} = \frac{1}{|R|} \sum_{r \in R} \max_{\hat{r} \in \hat{R}} \text{sim}(r, \hat{r}),$$

where sim denotes cosine similarity; F1 is the harmonic mean of these values. While reconstruction-based BERTScore captures semantic coverage between the graph and the source narrative, it does not explicitly detect hallucinated content or subtle clinical inaccuracies, motivating complementary human evaluation.

**Topology Validation.** Structural integrity of the patient trajectory graphs is sanity checked using a validation routine that audits desired topological properties. For each graph, we assess acyclicity through depth-first search. We compute the number of weakly connected components to detect disconnected subgraphs and extract summary statistics including node and edge counts, average in-degree, and graph density. These metrics provide quality control and enable downstream filtering of malformed or incomplete graphs, helping ensure that the output graphs preserve logical flow and are suitable for downstream trajectory modeling.

### 5.2. *Clustering of Patient Trajectories*

We evaluate patient trajectory representations in two modalities: (1) derived Monopath DAGs and (2) corresponding free-text cases. Both representations are embedded using a pretrained ClinicalBERT model to yield fixed-length vector embeddings. For graph-based representations, token-level embeddings from node text are aggregated using mean pooling. For text-based representations, the entire vignette is encoded directly. This approach enables direct comparison between graph-implicit segmentation and denoising and naive linguistic representations.

To prepare high-dimensional embeddings for clustering, we apply two complementary dimensionality reduction methods: (i) PCA with whitening to de-correlate dimensions and normalize variances, and (ii) UMAP with default hyperparameters (15 neighbors, minimum distance = 0.1, Euclidean metric), as recommended by McInnes *et al.*.[26] We then apply K-means clustering to the reduced embeddings, evaluating cluster configurations with $k \in [2, 10]$. This evaluates whether graph-induced normalization improves narrative case report representations. Clustering quality is assessed using the Calinski–Harabasz Index (CH) for both graph-based and text-based embeddings under three settings: raw, PCA-reduced, and UMAP-reduced. To examine clinical relevance, we incorporate structured metadata on binary metastasis status, computing the proportion of metastatic versus non-metastatic cases within each cluster and normalizing for cluster size to enable fair comparison.

We use the Calinski–Harabasz Index because it is scale-invariant, does not require labeled ground truth, and directly measures between-cluster separation relative to within-cluster compactness, making it suitable for unsupervised representation comparison.

### 5.3. *Synthetic Trajectories and Clinical Expert Review*

**Synthetic Generation.** For a random subset of cases, we generate two synthetic narratives per patient. In the text-only baseline, segments of the original case report are summarized independently and then recombined into a single narrative without access to explicit patient states or transitions. In the graph-conditioned setting, the narrative is generated from the ordered nodes and edges of the corresponding patient trajectory graph $G_i$, providing the model with explicit state and transition structure. Both conditions use the same underlying language model, allowing us to isolate the effect of trajectory structure on the generated narrative.

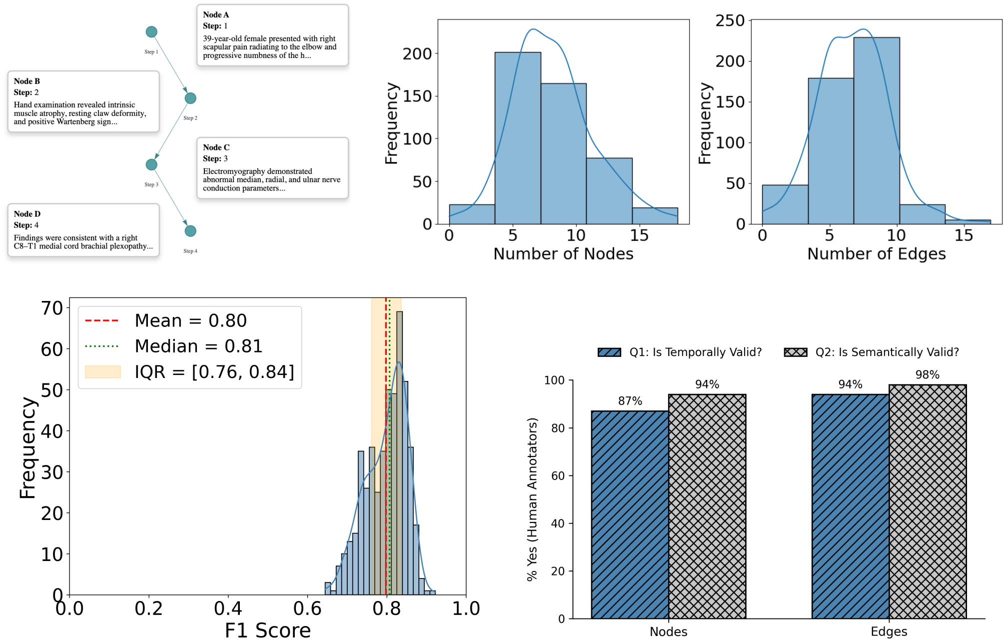
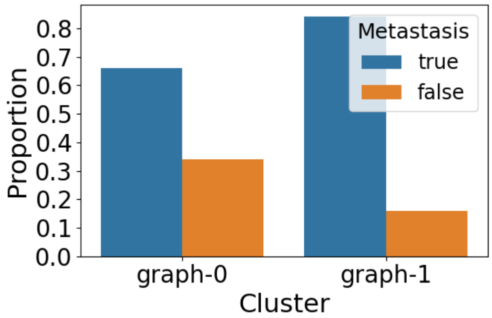



Fig. 2: **Monopath DAG trajectory representations and evaluation.** (a) Schematic of a monolithic DAG representing a patient case. (b) Distribution of node and edge counts across constructed graphs. (c) BERTScore F1 similarity between reconstructed narratives and original texts. (d) Human evaluation comparing graph fidelity and ordering accuracy.

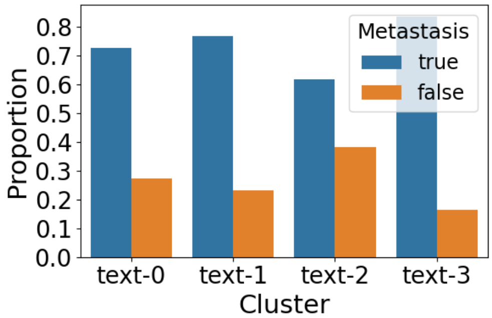
**Clinician-centered evaluation criteria**

**Q1. (Clinical validity):** Baseline check that case actions are medically sound.
**Q2. (Timeline validity):** Assesses whether the case progression follows a clinically appropriate order, capturing temporal reasoning.
**Q3. (Clinically actionable):** Measures whether the case is specific and granular enough to guide the next clinical decision.
**Q4. (Safely clinically actionable):** Measures whether the information is sufficient to make a safe decision with confidence, reflecting decision certainty.
**Q5. (Appropriate language):** Evaluates whether the narrative uses medical language appropriately, avoiding generic or ambiguous phrasing.

**Clinical Expert Review.** We conducted a single-blinded evaluation with four practicing physicians, who were shown a mix of real and synthetic patient cases and asked to rate each on a 5-point Likert scale for realism and clinical plausibility. To move beyond correctness as the sole benchmark, we designed a clinician-centered framework that reflects the auxiliary principles guiding real-world decision-making. Five targeted questions (Q1–Q5), reviewed by medical experts, captured hidden reasoning signals—validity, temporal orientation, specificity, safety, and trust in communication—that are absent from standard benchmarks but central to clinical impact. Q1 assesses medical correctness in isolation, whereas Q4 evaluates whether the information provided supports confident and safe decision-making under clinical uncertainty (Appendix 4.3).




(a)




(b)

Fig. 3: **Cluster-level metastasis composition from graph- and text-based embeddings.** (a) Graph-based embeddings show distinct enrichment patterns by metastasis status across clusters. (b) Text-based embeddings exhibit similar trends, though with more heterogeneous distributions.

## 6. Results and Discussion

From the initial set of 550 case reports, we remove non-English entries and case series, resulting in a curated dataset of 485 individual narratives. Semantic fidelity of the graph-derived reconstructions is evaluated using BERTScore, yielding an average BERTScore F1 of 0.798, median 0.807, and an interquartile range between [0.761, 0.836]. Scores span from 0.676 to 0.876, with low variance ($\sigma \approx 0.051$), pointing to stable reconstruction quality across cases. The lower F1 scores observed point to a subset of challenging cases with lower alignment. These outliers may reflect failures in content preservation or structural reconstruction. Overall, the high average scores and low variance suggest that the model reliably reproduces the core clinical information and narrative structure across diverse cases. Precision and recall components of BERTScore are 0.790 and 0.807, respectively (Appendix 2.1). On average, graphs contained $8\pm3$ nodes and $7\pm3$ edges, indicating clinically meaningful structural detail. Human evaluation of 568 nodes and edges further supports the structural consistency of the generated graphs. Node order is correct in 87% of cases, and 94% of nodes are deemed semantically accurate. For edges, correctness reaches 94%, and semantic validity approaches 98 %, supporting the effectiveness of the graph generation process in preserving temporal logic and clinical meaning

(Figure 2b).

Clustering analyses across raw, PCA-reduced, and UMAP-reduced embedding spaces consistently favor graph-based representations over text. In the raw space, graphs attain a Calinski-Harabasz score of 110.5, significantly exceeding the 41.9 achieved by text-based embeddings. We hypothesize that this improvement arises because graph-based representations normalize temporal structure and clinical state transitions, reducing lexical noise and emphasizing progression patterns that are diluted in free-text embeddings. Dimensionality reduction preserves this advantage: PCA boosts the score to 207.9 for graphs (*vs.* 80.9 for text), while UMAP accentuates it further—489.2 compared to 289.4. The structured encoding of clinical events within graphs appears to enhance cluster separability, capturing nuanced variation across patient trajectories. When stratifying patients by metastasis status, graph-based clusters display sharper separation. As illustrated in Figure 3, the graph-based clustering (left) yields three distinct groups. Cluster graph-0 and graph-2 show strong enrichment for metastatic cases, with over 80% of instances labeled as metastatic, while graph-1 presented a more mixed composition with approximately 65% metastatic and 34% non-metastatic cases. In contrast, the text-based clustering (right) reveals four clusters with more variable metastatic distributions. Clusters text-0, text-1, and text-2 each contain over 70% metastatic cases, with text-1 being the most enriched ($\approx$ 85%). However, text-3 shows the lowest metastatic proportion ($\approx$ 63%) and the highest fraction of non-metastatic cases among text clusters. These findings indicate that graph-based embeddings more distinctly stratify patients by metastasis status compared to text-based representations. These results indicate that graph-conditioned representations better surface clinically impactful signals, such as temporal coherence, specificity and safety.

A blinded evaluation by four physicians (PGY-1 to PGY-3) compared control-generated and graph-conditioned synthetic cases across the five clinical criteria (Figure 4, Table 1). PGY-level physicians were selected because they routinely perform frontline clinical reasoning and decision-making, making them well positioned to assess timeline coherence, actionability, and safety in narrative cases. Their evaluations reflect the perspective of clinicians who most frequently rely on concise, temporally coherent summaries in practice. Results highlight that while clinical validity (Q1) remained stable across conditions, graphs yielded significant improvements in timeline validity (Q2, $p = 0.003$), actionability (Q3, $p = 0.004$), and safe actionability (Q4, $p < 0.001$). These three dimensions are critical because they reflect the reasoning traces clinicians need: chronological progression, specificity for decision support, and confidence in safety. Language appropriateness (Q5) trended positive (p = 0.051), suggesting graphs may reduce generic or buzzword-heavy phrasing. Taken together, these results suggest that Monopath DAGs do not simply preserve correctness but enhance the qualities that make cases useful for clinical decision-making. Although the binary evaluation schema (Q1: order, Q2: accuracy) provides only a coarse-grained view of graph fidelity, it is designed to complement automated semantic metrics such as BERTScore with a minimal human check that could be performed within the constraints of clinician time. This design emphasized inter-reviewer agreement and feasibility, but it does not capture node informativeness or the clinical validity of edges on its own. We consider this a key limitation of the current evaluation and plan to

Table 1: Paired t-tests comparing control vs. sample responses across five Likert-scale questions. Negative $t$ and $d$ values indicate higher scores in the sample condition. Values are mean $\pm$ SD. ($^*p < 0.05$, $n = 106$ pairs).

| **Question** | **Control** | **Sample** | $t$ | $p$ | **Cohen's** $d$ |
|---|---|---|---|---|---|
| Clinical validity | $3.25 \pm 1.07$ | $3.44 \pm 1.05$ | $-1.37$ | 0.175 | $-0.14$ |
| Clinical timeline validity | $2.98 \pm 1.15$ | $3.43 \pm 1.04$ | $-3.05$ | 0.003* | $-0.31$ |
| Clinically actionable | $2.64 \pm 1.04$ | $3.10 \pm 1.04$ | $-2.96$ | 0.004* | $-0.30$ |
| Safely clinically actionable | $2.28 \pm 0.95$ | $2.75 \pm 0.92$ | $-3.59$ | $< 0.001$* | $-0.37$ |
| Appropriate clinical language | $3.28 \pm 1.08$ | $3.56 \pm 0.94$ | $-1.97$ | 0.051 | $-0.20$ |

extend our framework with more fine-grained protocols, including explicit node-level assessments and validation of transition plausibility. As noted in Section 4.3, clinician evaluation compared synthetic narratives with control narratives on textual criteria; physicians did not review graphs directly or express preferences between graph and text formats. In this study, graphs are used as structured intermediates to support generation and retrieval, rather than as interfaces intended for direct clinical decision-making.

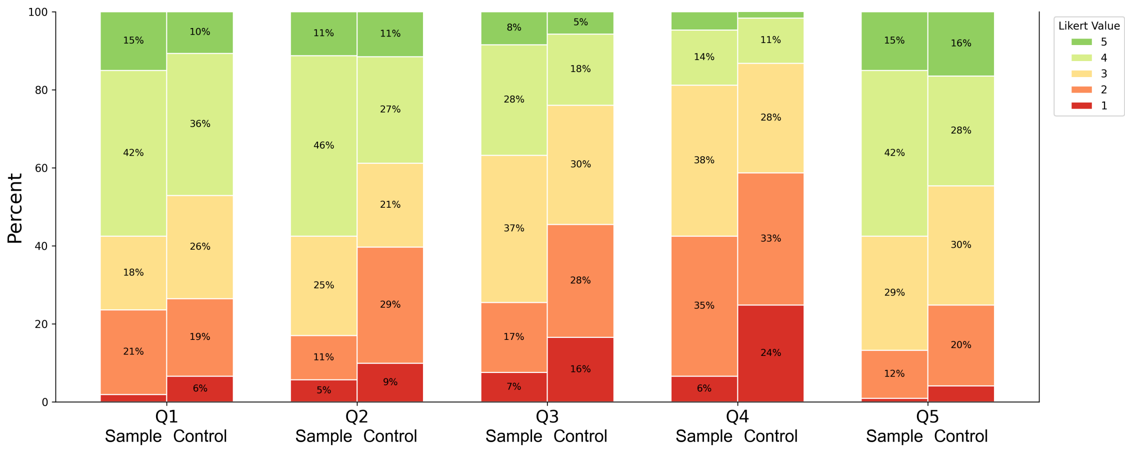



Fig. 4: Evaluation of synthetic narratives by human clinical experts.(n=218)

## 7. Generalizability across Conditions

While our primary experiments focus on lung cancer—chosen for its prevalence in case report literature, detailed longitudinal trajectories, and heterogeneous treatment paths—the pipeline is agnostic to disease. It operates directly on free-text narratives and standardizes outputs via UMLS, integrated into DSPy docstring prompts and JSON output. Ontology grounding ensures that extracted events can be normalized across conditions and datasets.

### 7.1. *Rare Diseases.*

We conducted additional experiments on four rare diseases across body systems (Gaucher, Wilson's, Charcot–Marie–Tooth, and Retinitis Pigmentosa), using 50 case reports per disease (200 total, see Appendix 2.2). These conditions were selected because they span distinct organ systems, disease progression patterns, and diagnostic pathways, providing a diverse test bed for evaluating generalization beyond oncology-specific narratives. Despite reduced sample size ($n = 200$ vs. 485) and minimal explicit clinical labels (*e.g.*, metastasis status, subtype), Monopath DAGs consistently improved clustering quality compared to text, with Calinski–Harabasz score gains of +21% (Raw), +24% (PCA), and +101% (UMAP). These findings reinforce the structural advantages of graph-based representations even under constrained conditions, and suggest that larger, labeled cohorts are likely to yield further improvements. While a full-scale evaluation across multiple diseases and languages is beyond the current study, these rare-disease results provide initial evidence of broader extensibility.

## 8. Conclusion and Future Work

We show that making the temporal and causal structure of clinical case reports explicit produces a reusable representation for trajectory comparison, clustering, and controlled generation. Monopath DAGs preserve source content while exposing rare and nuanced clinical paths that are difficult to study systematically in standard EHR data. These graphs support trajectory-aware case matching, edge-level counterfactual modification, and synthetic cohort generation, providing a structured intermediate for data augmentation, model stress-testing, and downstream analysis.

**Limitations.** Our evaluation is limited to recent, English-language lung cancer case reports and may not generalize across specialties, languages, or narrative conventions. Monopath DAGs enforce a single dominant clinical thread, collapsing concurrent events, branching trajectories, and cyclic complications. The extraction pipeline also depends on general-purpose LLMs and retains errors in negation, temporal ordering, and dosage extraction that are not fully captured by BERTScore or our limited human evaluation. Synthetic trajectories were evaluated through small-scale expert review, without privacy-leakage audits or downstream task benchmarking. Thus, the Monopath representation improves interpretability and temporal clarity at the cost of expressivity.

**Future Work.** We plan to extend Monopath DAGs to *Multipath DAGs* with controlled branching to represent clinical uncertainty, reversible events, and alternative pathways. We will also investigate biomedical language models and graph-imputation methods to improve extraction of under-specified clinical states. These extensions could support trajectory-informed outcome prediction, counterfactual synthetic generation, cross-disease benchmarking, and trajectory-aware case matching for clinical decision support.

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## 1. Notations and Definitions

### 1.1. *Task definition*

We focus on the monopath constraint—each node has at most one incoming edge—to establish feasibility of robust, large-scale extraction and to preserve temporal directionality. More expressive graph structures (*e.g.*, multiplex or hypergraphs) can capture concurrent or cyclic phenomena but complicate interpretability. We view them as complementary extensions (see Limitations).

$$\mathcal{D} = \{d_1, d_2, \ldots, d_N\} \tag{1}$$

denotes a corpus of free-text clinical case reports, where each $d_i$ describes a longitudinal patient narrative. For each report $d_i$, our objective is to construct a directed acyclic graph (DAG)

$$G_i = (V_i, E_i) \in \mathcal{G}, \qquad i = 1, \ldots, N, \tag{2}$$

that captures the patient's clinical trajectory in a structured format.

**Graph Representation.** We denote a generic DAG as $G = (V, E)$, where

$$V = \{v_1, \ldots, v_K\}, \qquad\qquad E \subseteq V \times V. \tag{3}$$

**Node Semantics.** Each node $v_k \in V$ corresponds to a temporally anchored patient state and is annotated with:

$$\begin{aligned}
&s\colon V \to \mathcal{T}, && s(v_k) = s_k && \text{(free-text content)},\\
&a\colon V \to \mathcal{A}, && a(v_k) = a_k && \text{(structured clinical data attributes)},\\
&t\colon V \to \mathbb{R}_{>0} \cup \{\varnothing\}, && t(v_k) = t_k && \text{(optional timestamp)}.
\end{aligned} \tag{4}$$

The attribute set $a_k$ typically includes symptoms, diagnoses, laboratory values, medications, and procedures.

**Edge Semantics.** Each edge $e_{ij} = (v_i \to v_j) \in E$ denotes a transition between two patient states. We define edge annotations as:

$$\begin{aligned}
&\tau\colon E \to \mathcal{T}, && \tau(e_{ij}) = \tau_{ij} && \text{(transition type)},\\
&\phi\colon E \to 2^{\mathcal{C}}, && \phi(e_{ij}) = \varphi_{ij} && \text{(triggering clinical entities)},
\end{aligned} \tag{5}$$

where $\mathcal{C}$ denotes a standardized clinical ontology [*e.g.*, Unified Medical Language System (UMLS)]. Thus, each $G_i$ yields a machine-readable, temporally ordered abstraction of the unstructured narrative in $d_i$.

**Event Standardization.** To convert free-text narratives into machine-actionable graph elements, all events are normalized into ontology-grounded fields. Sentences are decomposed into atomic facts, and the `NodeClinicalDataExtract` module maps each fact to Unified Medical Language System (UMLS) concepts, yielding a structured `clinical_data` dictionary per node. Edges are similarly represented as typed JSON `transition_event` objects specifying trigger, domain, change type, and ontology-mapped entities ($\phi(e) \subseteq \mathcal{C}$, where $\mathcal{C}$ is the ontology). Narrative strings are retained only as evidence, while ontology grounding enables programmatic querying, aggregation, and embedding-based analysis.

### 1.2. *Notation for patient trajectory graphs*

This table summarizes the notation for patient trajectory graphs, including case reports, DAGs, clinical state nodes, and directed transitions. Nodes may include free-text summaries, structured attributes, timestamps, and mapped clinical concepts, while edges capture labeled state transitions. Together, the notation provides a consistent framework for representing patient trajectories.

Table 2: Notation for patient trajectory graphs

| **Notation** | **Description** |
|---|---|
| $D = \{d_1, d_2, \ldots, d_N\}$ | Corpus of free-text clinical case reports |
| $d_i \in D$ | A single case report |
| $G_i = (V_i, E_i)$ | DAG representing the patient trajectory for case $d_i$ |
| $\mathcal{G}$ | Set of all patient trajectory graphs |
| $V = \{v_1, \ldots, v_K\}$ | Set of nodes representing clinical states |
| $E \subseteq V \times V$ | Set of directed edges between states |
| $v_k \in V$ | A single patient state node |
| $s_k \in \mathcal{T}$ | Free-text content of the patient state at $v_k$ |
| $a_k \in \mathcal{A}$ | Structured clinical data attributes at $v_k$ |
| $t_k \in \mathbb{R}^+ \cup \{\emptyset\}$ | Optional timestamp for node $v_k$ |
| $e_{ij} = (v_i \to v_j) \in E$ | Directed transition from $v_i$ to $v_j$ |
| $\tau_{ij} \in \mathcal{T}$ | Transition type label for edge $e_{ij}$ |
| $\phi_{ij} \subseteq \mathcal{C}$ | Triggering clinical entities mapped to ontology |
| $\mathcal{T}$ | Space of free-text summaries or transition types |
| $\mathcal{A}$ | Space of structured clinical attributes |
| $\mathcal{C}$ | Space of clinical concepts/entities |

## 2. Additional Data

### 2.1. *Structure and semantics of DAGs*

Semantic accuracy is assessed using BERTScore (mean ± standard deviation) across precision, recall, and F1. Topological correctness is evaluated through graph-theoretic metrics: the percentage of acyclic graphs, the number of weakly connected components, and average node-level statistics (node count, edge count, in-degree, and graph density). These metrics quantify both structural validity and interpretability of the generated DAGs.

| **Metric** | **Monopath DAG** |
|---|---|
| **BERTScore (mean ± std)** | |
| Precision | $0.790 \pm 0.060$ |
| Recall | $0.807 \pm 0.047$ |
| F1 Score | $0.798 \pm 0.051$ |
| **Topology Validation** | |
| Acyclic | 99.175 % |
| Weakly Connected Component | $1.436 \pm 1.252$ |
| Avg. Node Count | $8.182 \pm 3.139$ |
| Avg. Edge Count | $6.749 \pm 2.567$ |
| Avg. In-Degree | $0.820 \pm 0.108$ |
| Graph Density | $0.137 \pm 0.068$ |

### 2.2. *Rare disease clustering output*

Comparison of Graph vs. Text representations across embeddings. Values shown are (Graph, Text), with percent improvement of Graph over Text. Silhouette score measures how well-separated clusters are, with higher values indicating more coherent and distinct groupings. The Calinski–Harabasz (CH) index evaluates clustering compactness and separation, where larger values reflect tighter, well-defined clusters.

| **Dataset** | **Embedding** | **Silhouette** | | **Calinski–Harabasz** | |
|---|---|---|---|---|---|
| | | Values | % Δ | Values | % Δ |
| Rare Disease | Raw | (0.1340, 0.0978) | +37.0% | (5.12, 4.23) | +21.0% |
| Rare Disease | PCA | (0.2450, 0.1777) | +37.9% | (15.63, 12.56) | +24.4% |
| Rare Disease | UMAP | (0.4845, 0.3727) | +29.9% | (103.25, 51.44) | +100.8% |
| Synthea | Raw | (0.0832, 0.0725) | +14.8% | (3.87, 3.15) | +22.9% |
| Synthea | PCA | (0.1904, 0.1627) | +17.0% | (8.55, 6.97) | +22.7% |
| Synthea | UMAP | (0.3328, 0.2301) | +44.7% | (5.11, 4.62) | +10.6% |

## 3. Output Overview

### 3.1. *Node and edge attributes*

The following table lists the structured attributes used in our graph representation of patient trajectories. Nodes represent individual clinical states, while edges denote transitions between those states. Not every field is present in every case. Only available information from the narrative is populated.

| Node Attributes | Edge Attributes |
|---|---|
| **Node ID:** unique identifier (e.g., "A", "B") in order of appearance. | **Edge ID:** concatenation of source and target node IDs (e.g., "A→B"). |
| **Step Index:** index in the sequence. | |
| **Narrative Content:** descriptive text of the state. | **Narrative Content:** description of the event or change. |
| **Timestamp:** extracted if explicitly mentioned. | **Transition Event:** structured encoding with: |
| **Clinical Data:** structured key-value fields (e.g., vitals, labs, medications, imaging, HPI, etc.). | `Trigger Type`, `Trigger Entities`, `Change Type`, `Affected Domain` |

### 3.2. *Node output example*

The following is an example of structured content for a node extracted from a patient case report. It includes narrative summary, structured clinical data (such as HPI and social history), and concept codes.

//

**Node Details:** `graph_001`

[left=1.5em, label=•]

- **id:** N1
- **label:** Step 1
- **customData:**
  [label=–]
  - **node_id:** A
  - **node_step_index:** 0
  - **content:** 44-year-old male presented with right-sided chest pain, dry cough, on-off fever, and hematuria for 2 months. Patient has no history of Antitubercular treatment (ATT) intake and was a tobacco chewer for >20 years. Physical examination revealed decreased air entry on the right side of the lung.
  - **clinical_data:**
    [label=–]
    - **HPI:**
      [label=–]
      - **summary:** right-sided chest pain, dry cough, on-off fever, and hematuria for 2 months
      - **duration:** 2 months
      - **associated_symptoms:** [label=∘]
      - C0008031
      - C0010200
      - C0015967
      - C0019062
    - **social_history:**
      [label=–]
      - **category:** tobacco
      - **status:** current
      - **description:** tobacco chewer for ¿20 years
- **custom_id:** graph_001_N0

### 3.3. *Edge output example*

The following is an example of a structured edge output from a case, representing a transition between two clinical states. It includes both narrative summary and structured transition fields.

//

**Edge Details:** `graph_001`

[left=1.5em, label=•]

- **from:** N1
- **to:** N2
- **data:**
  [label=–]
  - **edge_id:** A_to_B
  - **content:** Initial presentation and subsequent imaging
  - **transition_event:**
    [label=–]
    * **trigger_type:** imaging
    * **trigger_entities:**
    * **change_type:** addition
    * **target_domain:** imaging
- **custom_id:** graph_001_N1_N2
- **id:** 413a3142-63b0-4eeb-9830-26011a516e1a

### 3.4. *DAG construction algorithm*

To illustrate how free-text case reports are systematically transformed into structured trajectory graphs, we provide the procedure below. The algorithm outlines the steps for segmenting narratives, constructing nodes and edges, extracting clinical data, and serializing the resulting Monopath DAG representation.

**Algorithm 1.** [1]

**Input:** Free-text case report $d_i$ **Output:** Monopath DAG $G_i = (V_i, E_i)$

$p_{1:k} \leftarrow$ SEGMENT($d_i$, *window* = 10 *sentences*) $T_i \leftarrow$ PATIENTTIMELINE($p_{1:k}$) $V_i \leftarrow [\,]$

each block $c_j$ in CHUNK($T_i$, 4 *sentences*) $v \leftarrow$ NODECONSTRUCT($c_j$, mem $= V_i$) $s^{\text{atom}} \leftarrow$ DECOMPOSETOATOMICSENTENCES($v.s$) $v.a \leftarrow$ NODECLINICALDATAEXTRACT($s^{\text{atom}}$) $V_i \leftarrow$ MERGEIFDUPLICATE($v, V_i$)

$E_i \leftarrow [\,]$ $t = 1$ $|V_i| - 1$ $e \leftarrow$ EDGECONSTRUCT($V_i[t], V_i[t+1]$) append $e$ to $E_i$

$G_i \leftarrow$ SERIALIZEASJSON($V_i, E_i$, UMLS metadata) **return** $G_i$

## 4. Human Evaluation

### 4.1. *Human evaluation schematic*

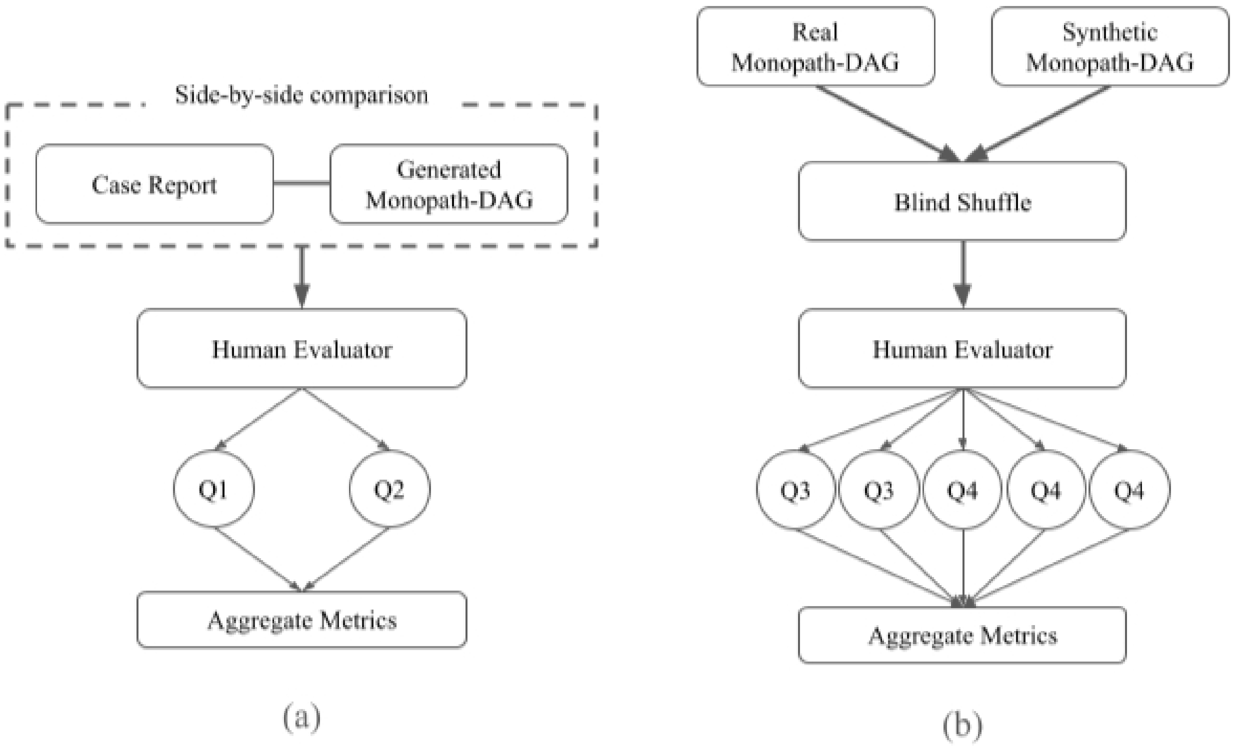


Fig. 5: Human evaluation schematics for (a) fidelity and (b) synthetic graph assessments.

### 4.2. *Fidelity Evaluation*

To assess whether generated Monopath-DAGs faithfully represent the underlying case reports, we implemented a two-part fidelity evaluation. First, we reconstructed narratives from graphs and compared them to original reports with BERTScore (ClinicalBERT embeddings). This provided automated precision, recall, and F1 measures of semantic similarity. Second, human annotators performed side-by-side comparisons of case reports and corresponding graphs, answering the following questions:

- **Q1:** Are the nodes and edges in the correct order? [Yes / No]
- **Q2:** Is the content of the graph accurate to the case report? [Yes / No]

Aggregate metrics from these assessments quantify structural and semantic fidelity (Figure 5a).

### 4.3. *Clinical Evaluation*

To move beyond correctness, we conducted a clinician-centered evaluation focused on decision-making relevance. Real and synthetic Monopath-DAGs were blind-shuffled and presented to practicing physicians, who rated each on five targeted criteria:

- **Q1 (Clinical validity):** Are the case actions and insights medically sound?
- **Q2 (Timeline validity):** Does the case progression follow a clinically appropriate order?
- **Q3 (Clinically actionable):** Is the case specific and granular enough to guide the next clinical decision?
- **Q4 (Safely clinically actionable):** Is the information sufficient to make a safe decision with confidence?
- **Q5 (Appropriate language):** Does the narrative use medical language appropriately, avoiding generic or ambiguous phrasing?

These criteria capture reasoning dimensions absent from standard benchmarks but essential for clinical impact. Aggregate metrics from these assessments provide a structured view of graph quality in real-world settings.

## 5. Prompt Design

### 5.1. *DAG primer*

Docstring used to prime the LLM for DAG extraction.

**Docstring:** `dag_primer`

```
You are an assistant that converts clinical case narratives into dynamic
Directed Acyclic Graphs (DAGs).

Each DAG consists of:
- Nodes = snapshots of the patient’s state.
- Edges = transitions between those states.

Terminology guidance:
- Use UMLS-standard concepts when possible for consistency and
interoperability.
- If a concept isn’t covered by UMLS, use clear, logical labeling.

Text extraction guidance:
- When looking at the case report input, ignore the references, background,
conclusions etc. sections
- We only want to extract on content relating to the specific patient
discussed in the case report
```

### 5.2. *Node instructions*

Instructions used to guide LLM-based node construction.

**Docstring:** `node_instructions`

```
You are given a clinical case report. Your task is to extract a sequence of
nodes representing the patient's evolving clinical state.

Guidelines:
- Create one node per clinically meaningful state.
- Combine co-occurring labs/imaging into the same node.
- Use separate nodes for clearly sequential or distinct events.
- Do not return anything outside the list format. Should be in JSON
compatible style.
- Keep imaging content packaged in one node if no clear temporal change is
indicated.
- Keep pathology / histology content packaged in one node if no clear
temporal change is indicated.
- node_memory is a running memory that updates as we add new nodes; use it to
preserve context, merge overlapping details, and avoid redundant or stale
states.

Output format:
Return a list of node dictionaries, in this order from top to bottom, each
with:
- node_id (In ascending alphabetical order, e.g., "A", "B", "C")
- node_step_index (integer for order)
- content (exhaustive clinical content, include all relevant details for the
given node)
- timestamp (optional, ISO8601)

Example:
{
"node_id": "A",
"node_step_index": 0,
"content": "The patient presented with bilateral painless testicular masses."
}
```

### 5.3. *Edge instructions*

Instructions used to guide LLM-based edge construction.

**Docstring:** `edge_instructions`

```
Each edge represents a change from one node to another. There should be one
edge in between every pair of adjacent nodes.

Guidelines for edges:
- Create edges only when there is a clear clinical progression or change
between nodes.
- Maintain narrative or logical order | edges should flow from earlier to
later events.
- Combine co-occurring findings into the same node, not across multiple
edges.

Output format:
Return a list of edge dictionaries, in this order, each with:
- edge_id: Unique identifier (Use format "node_id"_to_"node_id", such that the
first "node_id" is the upstream node and the second "node_id" is the
downstream node bounding the edge)
- content: Exhuastive clinical content, include all relevant details for the
given node

Optional structured field for edge-level transitions:
transition_event = {
   "trigger_type": "procedure | lab_change | medication_change | symptom_onset
| interpretation | spontaneous",
   "trigger_entities": ["UMLS_CUI_1", "UMLS_CUI_2"],        # e.g., C0025598 =
Metformin, C0011581 = Chest Pain
   "change_type": "addition | discontinuation | escalation | deescalation |
reinterpretation | resolution | progression | other",
   "target_domain": "medication | symptom | diagnosis | lab | imaging |
procedure | functional_status | vital_sign",
   "timestamp": "ISO 8601 datetime (e.g. , 2025-03-01T10:00:00Z"), only
include if explicitly given and can be converted to datetime"
}
```

### 5.4. *Node clinical data instructions*

Instructions for extracting structured clinical data per node.

**Docstring:** `node_clinical_data_instructions`

```
Each node includes an optional structured field clinical_data, formatted as a
dictionary with categories mapping to lists of dictionaries. Values should
only be included if matched to the UMLS Metathesaurus; otherwise, omit and do
not print/store it.

clinical_data = {
  "medications": [{
    "drug": "UMLS_CUI or string",
    "dosage": "string",
    "frequency": "string",
    "modality": "oral | IV | IM | subcutaneous | transdermal | inhaled |
other",
    "start_date": "ISO8601",
    "end_date": "ISO8601 or null",
    "indication": "UMLS_CUI or string"
  }],
  "vitals": [{...}],
  "labs": [{...}],
  "imaging": [{...}],
  "procedures": [{...}],
  "HPI": [{...}],
  "ROS": [{...}],
  "functional_status": [{...}],
  "mental_status": [{...}],
  "social_history": [{...}],
  "allergies": [{...}],
  "diagnoses": [{...}]
}
```

## 6. Additional Related Work

### 6.1. *Expansion beyond existing extraction methods*

We situate Monopath DAGs within prior efforts on case report extraction and patient-journey modeling. The below table summarizes representative approaches, contrasting pipelines, outputs, and downstream utility. Whereas earlier resources focus on entity spans, citation heuristics, or free-text knowledge graphs, our method provides temporally ordered, ontology-grounded DAGs that directly capture the structure of patient trajectories.

| Work | Extraction Pipeline | Output | Utility |
|---|---|---|---|
| **Monopath DAGs** | LLM + DSPy pipeline | UMLS-grounded DAG per patient | Supports trajectory retrieval, counterfactual edits, synthetic cohorts; clinically validated |
| **PMC-Patients** (Zhao, 2023) | Regex & heuristics; citation links | Paragraph + triples | Retrieval benchmark only; no temporal or causal |
| **CaseReportBench** (Zhang, 2025) | Human BIO annotation | Entity spans | High-quality labels; no temporal or relation structure |
| **Patient-Journey KGs** (Al Khatib, 2025) | Prompt-based LLM; node merging | Multi-relational KG per patient | Structural compliance evaluation; no clinical validation |

**Comparison with PMC-Patients.** While both our work and Zhao et al. (PMC-Patients) leverage case reports, the approaches differ fundamentally.

*Scope:* PMC-Patients is designed as a retrieval benchmark using free-text patient summaries and citation links, whereas our work extracts structured representations and models causal dependencies across events.
*Granularity:* PMC-Patients preserves paragraphs, but Monopath DAGs break narratives into machine-readable node–edge pairs, annotated with timestamps and UMLS terms, enabling clustering, simulation, and downstream reasoning.
*Evaluation:* Zhao et al. report retrieval performance; we instead validate semantic fidelity (BERTScore), demonstrate improved trajectory clustering (Calinski–Harabasz), and conduct blinded physician assessments of synthetic realism.
*Use-case:* By encoding causal flow, DAGs enable applications beyond retrieval, including trajectory-aware search, counterfactual simulation, and synthetic cohort generation.

## 7. Computational Resources

The pipeline is implemented in Python and runs on standard research hardware. Experiments used a single NVIDIA A100 GPU (40GB) with 64GB RAM, though CPU-only execution is possible with longer runtimes. End-to-end processing takes about 2–3 minutes per case report, making the approach accessible to academic and clinical groups.